Samy Derradji

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Algiers, Algeria

PROFESSIONAL SUMMARY

Electronic and embedded systems engineer with **3+ years** shipping IoT/robotics hardware **end-to-end** - schematic capture, PCB layout, board bring-up, and firmware on **STM32/ESP32** - plus doctoral research in **low-power RISC-V SoC** design (FPGA/ASIC, SystemVerilog verification). Full hardware-lifecycle owner, fluent in French and English, prioritising embedded and digital-design roles in France.

CORE COMPETENCIES

Embedded Firmware (C/C++, Rust, MicroPython) | PCB Design (KiCad, EasyEDA, Altium) | STM32 / ESP32 / AVR MCUs | VHDL & SystemVerilog, FPGA (Zynq-7000) | UART | I²C | SPI | CAN | USB | RS485 | Wi-Fi | BLE | ESP-NOW | Board Bring-up & Lab Debugging | Low-Power RISC-V SoC (QoSoC)

WORK EXPERIENCE

Techmology

Sept 2023 - Present

Embedded Systems Engineer (Hardware & Firmware)

Algiers, Algeria

- Designed and shipped educational robotics / IoT hardware products end-to-end: schematic capture, PCB layout, component selection/BOM, and embedded firmware.
- Developed firmware for STM32 and other MCUs covering motor control, sensor acquisition, and communication interfaces (UART, I²C, SPI, CAN).
- Performed PCB assembly, board bring-up, and hardware debugging using lab equipment (oscilloscope, multimeter, logic analyzer).
- Carried out product testing and verification across multiple finished boards, from prototype through production.

Univ. of Blida 1 / CDTA

2025 - Present

Doctoral Researcher - Low-Power CPU/SoC Design (FPGA/ASIC)

Algiers, Algeria

- Core contributor to QoSoC, an open-source low-power RISC-V SoC fabricated on the IHP SG13G2 130nm PDK.
- Led FPGA-level functional verification in SystemVerilog and RTL/Wishbone integration debug.

Techmology

Aug 2022

Embedded Systems Intern

Algiers, Algeria

- Contributed to a smart IoT greenhouse system from sensor data capture to visualization.
- Developed ESP32 firmware applying low-power techniques.

PROJECTS

MCU2972 - QoSoC 32-bit RISC-V MCU

Taped out | Apache 2.0

Open-source 32-bit RISC-V MCU (RV32EC) for low-power IoT / embedded control, co-developed with the CDTA team and taped out on the IHP SG13G2 130nm PDK. Contributions: digital design, FPGA-level functional verification (SystemVerilog), RTL/Wishbone debug. PicoRV32 core, QSPI XIP (up to 8 MB external flash), 2 KB SRAM, UART/I²C/GPIO, timer with PWM/capture, dual RTC, watchdog, 5-state PMU with per-peripheral clock gating. Clean LVS/DRC/antenna, no setup/hold violations; ~38k standard cells, ~5.2 mW @ 16 MHz.

Verilog, PicoRV32, IHP SG13G2 130nm

Edge AI Human Detection System Master's thesis | 2023

Custom ESP32-S3 + OV2640 PCB (EasyEDA, JLCPCB) with 3D-printed enclosure running a quantized TensorFlow Lite Micro model for on-device human detection - inference entirely on the microcontroller, no cloud dependency.

ESP32-S3, OV2640, TensorFlow Lite Micro

ESP32 STEM Robotics Controller & Custom Firmware Shipping product

ESP32 PCB (6×AAA) driving motors and on-board sensors (line, color, compass, ultrasonic); custom MicroPython image with C driver bindings and a BLE bridge layer for Scratch compatibility.

ESP32, Custom MicroPython, BLE

ESP32-S3 IoT Sensor Board & Custom Firmware Shipping product

ESP32-S3 PCB (USB or 6×AAA) driving six sensor types (gas, LDR, DHT, PIR, RFID...) and four outputs (OLED, 7-segment, servo, 5V fan), built on the same custom MicroPython image and driver bindings.

ESP32-S3, Wi-Fi/BLE, Custom MicroPython

STM32 4-Wheeled Robot Motor Control Board

2-layer PCB controlling four motors independently via hall-encoder feedback; firmware in Embedded C; UART, CAN and USB interfaces with I²C sensor expansion.

STM32F103, Embedded C, PCB layout

Soil Quality Measurement System

2-layer PCB for remote soil monitoring with GSM uplink; optimized power delivery and signal integrity for stable long-distance data acquisition.

Atmega328P, SIM800L, RS485, Altium

16-bit Basic CPU in VHDL

Implemented and simulated a 16-bit CPU, validated on the Zynq Z7-10 FPGA; wrote a Python compiler from assembly to memory map.

VHDL, Xilinx Zynq Z7-10, Python

IoT Educational Platform

2-layer PCB integrating sensors and actuators; scratch-compatible MicroPython firmware with an embedded micro server; Tauri desktop client (Rust + Svelte) over Wi-Fi.

MicroPython, Rust, Svelte, Tauri

EDUCATION

PhD, Microelectronics / Low-Power SoC Design Univ. of Blida 1 / CDTA 2025 - Present

Algiers, Algeria

Low-power CPU architectures, VLSI design, SoC, open-source workflows (OpenROAD, Xschem).

M.S., Embedded Systems Electronics USTHB 2023

Algiers, Algeria

Final-year project: Edge AI person detection camera.

B.S., Electronics USTHB 2021

Algiers, Algeria

Final-year project: Smart IoT windvane and anemometer.

SKILLS

Programming: C++, Embedded C, Rust, MATLAB, Python, MicroPython, VHDL, SystemVerilog

Microcontrollers: STM32F103, ESP32, ESP32-S3, Atmega328P; ESP-IDF, STM32Cube, custom MicroPython builds & drivers

Electronics & PCB: Schematic capture, 2-layer PCB layout (KiCad, EasyEDA, Altium), BOM management, SMT soldering & bring-up

Communication / Wireless: UART, I²C, SPI, CAN, RS485, USB; Wi-Fi, BLE, ESP-NOW, MQTT

Test & Debug: Oscilloscope, multimeter, logic analyzer, hardware probing, fault analysis, product verification & QA

Digital Design: VHDL, Verilog, FPGA (Xilinx Zynq-7000), CPU architecture **IC Design:** OpenROAD flow scripts, Xschem, ngspice

Robotics & Software: ROS2 / Nav2, PX4 / Gazebo SITL, MQTT **Languages:** Arabic (native), French (fluent), English (professional)